



Chitkara

PAPER OF CSE-2025 - PYTHON TEST3 EXAM

Q1. Incremented Salary calculator

An organization made a policy for annual appraisals for the employees.

According to the policy, if the gender of the employee is female then 15% of the current gross salary or if the gender of the employee

is male then 10% of the current gross salary is added in the current gross salary of the employee to generate his/her new gross salary.

Write a program in python which can accept the input as the basic salary of the employee along with his/her current gross salary and the gender of the employee as 'M' for male or 'F' for Female.

Then the program will print the new Gross salary of the employee as the sum of current gross salary and increment (10% or 15%) based on gender. In case of wrong gender information, the program will print " wrong input"

Input Format:

First line of input contains the current gross salary of the employee

Second line of input contains the gender information of employee as "M" or "F" for male and female respectively.

Output Format:

Print new gross salary of the employee

Example 1

Input

1000
F

Output

1150.0

Example 2

Input

1000
k

Output

wrong input

ANSWER:

Test case 1 :

Input:
1000

k

Output:
wrong input

Test case 2 :

Input:
1000
F

Output:
1150.0

Test case 3 :

Input:
12000
Z

Output:
wrong input

Test case 4 :

Input:
15000
M
Output:
16500.0

Test case 5 :

Input:
20000
F

Output:
23000.0

Test case 6 :

Input:
100000
F

Output:
115000.0

Test case 7 :

Input:
1000000
M

Output:
1100000.0

Q2. Library Charges Problem

In some foreign university, the central library has charged a fine for every book returned late after the due date mentioned on the book. For first 5 days after due date the fine is 50 paise per day, if days are in between 6-10 days then the fine is one rupee per day and if days are in between 11-30 days then the fine is 5 rupees per day. If you return the book after 30 days

your membership will be cancelled. Write a program to accept the number of days the member is late to return the book and display the fine or the appropriate message.

Number of Days Fine per day

1-5 days 50 paisa per day

6-10 days 1 Rs per day

11-30 days 5 Rs per day

More than 30 days Your Membership would be Cancelled

Input Format:

Enter the number of days the member late to return book. (Integer value)

Output Format:

Print Message:

Fine Amount to Pay Rs: (If number of days are less than equal to 30)

Your Membership would be Cancelled (If number of days are grater than equal to 30)

Example 1

Input

4

Output

Fine Amount to Pay Rs: 2

Example 2

Input

50

Output

Your Membership would be Cancelled

ANSWER:Test case 1 :

Input:

50

Output:

Your Membership would be Cancelled

Test case 2 :

Input:

6

Output:

Fine Amount to Pay Rs: 6

Test case 3 :

Input:

32

Output:

Your Membership would be Cancelled

Test case 4 :

Input:

13

Output:

Fine Amount to Pay Rs: 65

Test case 5 :

Input:

20

Output:

Fine Amount to Pay Rs: 100

Test case 6 :

Input:

16

Output:

Fine Amount to Pay Rs: 80

Test case 7 :

Input:

28

Output:

Fine Amount to Pay Rs: 140

Q3. Leap Year

Once upon a time, a curious student Manu sat in front of their computer, eager to learn about leap years. "Is this year a leap year?" they wondered aloud. The wise computer, always ready with an answer, replied with a simple rule. "A year is a leap year if it's divisible by 4, but not by 100, unless it's also divisible by 400." Intrigued, the student thought deeply about the rule, repeating it to themselves. They realized how clever this logic was and felt proud of the newfound knowledge. From that day on, they could easily identify a leap year with confidence.

INPUT:

The program accepts a single integer input from the user, which represents the year to be checked.

OUTPUT:

The program outputs a message indicating whether the given year is a leap year or not.

Example :

Input: 2050

Output: 2050 is NOT a Leap Year.

ANSWER:Test case 1 :

Input:

2000

Output:
2000 is a Leap Year.

Test case 2 :

Input:
1900
Output:
1900 is NOT a Leap Year.

Test case 3 :

Input:
2024
Output:
2024 is a Leap Year.

Test case 4 :

Input:
2023
Output:
2023 is NOT a Leap Year.

Test case 5 :

Input:
2100
Output:
2100 is NOT a Leap Year.

Test case 6 :

Input:
2400
Output:
2400 is a Leap Year.

Q4. Matrix Border Summation

Problem Statement

You are given a matrix of size $N \times M$. Your task is to calculate the sum of the elements on the borders of the matrix. The border elements are those that are in the first row, last row, first column, and last column of the matrix.

Input Format:

The first line contains two integers N and M , representing the number of rows and columns in the matrix. The next N lines contain M integers each, representing the elements of the matrix.

Output Format:

Print a single integer, the sum of the border elements of the matrix.

Code Constraints:

$2 \leq N, M \leq 100$
Matrix elements are integers between -1000 and 1000 .

Sample Testcase 1

Input:

```
4 4
3 5 6 7
1 3 5 7
7 6 3 1
9 7 8 6
```

Output:

67

Explanation:

The border elements are 3,5,6,7,7,1,6,8,7,9,7,1 and their sum is 67.

ANSWER:Test case 1 :

Input:

```
4 4
3 5 6 7
1 3 5 7
7 6 3 1
9 7 8 6
```

Output:

67

Test case 2 :

Input:

```
3 3
2 2 2
2 2 2
2 2 2
```

Output:

16

Test case 3 :

Input:

```
2 4
1 2 3 4
5 6 7 8
```

Output:

36

Test case 4 :

Input:

```
1 5
1 2 3 4 5
```

Output:

15

Test case 5 :

Input:

```
4 2
1 45
2 32
3 20
```

```
4 11
Output:
118
```

Test case 6 :

```
Input:
5 5
1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
Output:
208
```

Test case 7 :

```
Input:
3 3
-1 -2 -3
-4 -5 -6
-7 -8 -9
Output:
-40
```

Q5. Row-wise Maximums

Problem Statement

In a special grid called a matrix, each number lives in a box. This grid has rows (like layers) and columns (like columns in a building).

Your job is to find the biggest number in each row of the grid. After you find these biggest numbers, you need to put them together in a list and show it.

Input Format:

The first line contains number of rows represented by `r`
The second line contains number of columns represented by `c`
The next `r` lines will each have `c` numbers (separated by space), which are the numbers in the grid.

Output Format:

The output prints a list of the biggest numbers from each row.

Code Constraints:

$1 \leq r, c \leq 50$,
 $0 \leq \text{matrix}[i][j] \leq 1000$

Sample Testcase 1

Input:

```
3
```

```
4
1 2 3 12
10 6 7 8
9 5 11 4
```

Output:

```
[12, 10, 11]
```

ANSWER:

Test case 1 :

Input:

```
3
4
1 2 3 12
10 6 7 8
9 5 11 4
```

Output:

```
[12, 10, 11]
```

Test case 2 :

Input:

```
3
4
7 7 7 7
7 7 7 7
7 7 7 7
```

Output:

```
[7, 7, 7]
```

Test case 3 :

Input:

```
3
2
6 4
5 9
8 6
```

Output:

```
[6, 9, 8]
```

Test case 4 :

Input:

```
4
1
100
200
300
400
```

Output:

```
[100, 200, 300, 400]
```

Test case 5 :

Input:

```
2
2
11 20
30 21
```

Output:

```
[20, 30]
```


Test case 6 :

Input:

2
1
4
5

Output:

[4, 5]

Q6. Column-wise Maximums**Problem Statement**

In a special grid called a matrix, each number lives in a box. This grid has rows (like layers) and columns (like columns in a building).

Your job is to find the biggest number in each column of the grid. After you find these biggest numbers, you need to put them together in a list and show it.

Input Format:

The first line contains number of rows represented by r

The second line contains number of columns represented by c

The next r lines will each have c numbers (separated by space), which are the numbers in the grid.

Output Format:

The output prints a list of the biggest numbers from each column.

Code Constraints:

$1 \leq r, c \leq 50$,
 $0 \leq \text{matrix}[i][j] \leq 1000$

Sample Testcase 1

Input:

3
4
1 2 3 12
10 6 7 8
9 5 11 4

Output:

[10, 6, 11, 12]

ANSWER:Test case 1 :

Input:

3
4

1 2 3 12
10 6 7 8
9 5 11 4
Output:
[10, 6, 11, 12]

Test case 2 :

Input:
2
3
5 3 1
4 7 8
Output:
[5, 7, 8]

Test case 3 :

Input:
3
4
7 7 7 7
7 7 7 7
7 7 7 7
Output:
[7, 7, 7, 7]

Test case 4 :

Input:
3
2
6 4
5 9
8 6
Output:
[8, 9]

Test case 5 :

Input:
4
1
100
200
300
400
Output:
[400]

Test case 6 :

Input:
2
2
11 20
30 21
Output:
[30, 21]

Test case 7 :

Input:
2
1
4
5

Output:
[5]

Q7. String is Palindrome or Not

As part of their practice in string handling, students have been asked to write a program that checks if a given word or phrase is a palindrome. A palindrome is a string that reads the same backward as forward. The program should take an input string and determine whether it is a palindrome.

Sample Test cases

Sample Input 1

Hello

Sample Output 1

Hello is not a palindrome

Sample Input 2

madam

Sample Output 2

madam is a palindrome

ANSWER:

Test case 1 :

Input:
madam

Output:
madam is a palindrome

Test case 2 :

Input:
Hello

Output:
Hello is not a palindrome

Test case 3 :

Input:
I Love Chitkara

Output:
I Love Chitkara is not a palindrome

Test case 4 :

Input:
racecar

Output:
racecar is a palindrome

Test case 5 :

Input:
radar

Output:
radar is a palindrome

Test case 6 :

Input:
Chitkara University

Output:
Chitkara University is not a palindrome

Test case 7 :

Input:
I study in Chitkara
Output:
I study in Chitkara is not a palindrome

Q8. Case Transition Count

Write a program that counts the number of positions where an uppercase letter immediately follows a lowercase letter in a given string. The program should traverse the string, compare adjacent characters, and count the occurrences where a lowercase letter is followed by an uppercase letter.

Program Requirements:

Input Validation:

Prompt the user to input a string.

Ensure the string contains at least two characters for meaningful comparison.

Counting Logic:

Traverse the string, checking if a lowercase letter is followed by an uppercase letter.

Count the number of such occurrences in the string.

Output:

Print the count of positions where an uppercase letter follows a lowercase letter.

Example 1:

Input:
helloWorld

Output:

1

Explanation:

- The string "helloWorld" has one position where a lowercase letter ("o") is followed by an uppercase letter ("W").

Example 2:

Input:

abCdEfGh

Output:

3

Explanation:

- The string "abCdEfGh" has three positions:
 - "bC"
 - "dE"
 - "fG"

Thus, the output is 3.

ANSWER:

Test case 1 :

Input:

helloWorld

Output:

1

Test case 2 :

Input:

aAaAa

Output:

2

Test case 3 :

Input:

lowerCASE

Output:

1

Test case 4 :

Input:

NoCasesHere

Output:

2

Test case 5 :

Input:

abcDEF

Output:

1

Test case 6 :

Input:

abCdEfGh

Output:

3

Test case 7 :

Input:

IndiaAAAA

Output:

1

Q9. Append new string in the middle of a given string

You are part of a development team for a social media platform that allows users to create customized captions for their posts. To make the captions more dynamic, users can provide two inputs: a primary message (string1) and an additional phrase (string2). Your task is to write a Python program that takes these inputs and seamlessly integrates the additional phrase into the middle of the primary message. This feature aims to help users craft engaging and personalized captions effortlessly.

Input:

First line contains a string as string1. ($1 \leq \text{len}(s1) \leq 1000$) for e.g; rohit

Second line contains a string string2. ($1 \leq \text{len}(s2) \leq 1000$) for e.g; singh

Output:

A string string3 which is formed by inserting string2 in the middle of string1. for e.g; rosinghhit

Sample Input :

hello
beautiful

Sample Output:

hebeautifullo

ANSWER:Test case 1 :

Input:

hello
beautiful

Output:

hebeautifullo

Test case 2 :

Input:

abcdef
XYZ

Output:

abcXYZdef

Test case 3 :

Input:

java

rocks
Output:
jarocksva

Test case 4 :

Input:
rahul
singh
Output:
rasinghhul

Test case 5 :

Input:
abc
x
Output:
axbc

Test case 6 :

Input:
carrace
fast
Output:
carfastrace

Test case 7 :

Input:
abcdefgh
123
Output:
abcd123efgh